AMRITA VISHWA VIDYAPEETHAM

Mathematics for Computing - 3

Optimization of Sparse Underwater Acoustic Channels using Robust ADMM Algorithm

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Introduction

- Underwater Acoustic (UWA) communication systems use sound waves to transmit data through water.
- These channels suffer from severe multipath propagation, large delay spreads, and impulsive noise.
- Accurate channel estimation is critical for coherent detection and reliable data transfer.
- Traditional methods perform poorly under impulsive noise, hence a robust ADMM-based approach is proposed.

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{n}$$

Problem Definition

The received signal model is:

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{n}$$

where:

- y: received signal vector (M × 1)
- A: known training/pilot matrix (M × N)
- x: channel coefficients to estimate (N × 1)
- n: noise vector (impulsive + Gaussian)The system is underdetermined (M < N), but x is sparse, enabling compressed sensing.

Optimization Formulation

The robust estimation problem is modeled as:

$$\min_{\mathbf{x}} \tau ||\mathbf{y} - \mathbf{A}\mathbf{x}||_1 + ||\mathbf{x}||_1$$

- First term: handles impulsive noise using L1 loss.
- Second term: promotes sparsity.
- τ: regularization parameter controlling the trade-off.

ADMM Formulation

Introduce auxiliary variable z = y - Ax:

$$\min_{\mathbf{x},\mathbf{z}} \ \tau||\mathbf{z}||_1 + ||\mathbf{x}||_1$$
$$\text{subject to } \mathbf{A}\mathbf{x} + \mathbf{z} = \mathbf{y}$$

$$L_\rho(\mathbf{x},\mathbf{z},\boldsymbol{\gamma}) = ||\mathbf{x}||_1 + \tau||\mathbf{z}||_1 + \frac{\rho}{2}||\mathbf{z}+\mathbf{A}\mathbf{x}-\mathbf{y}+\boldsymbol{\gamma}/\rho||_2^2 - \frac{1}{2\rho}||\boldsymbol{\gamma}||_2^2, \quad (5)$$

- x: main variable (channel estimate)
- z: residual term
- γ: Lagrange multiplier
- ρ: penalty parameter controlling convergence

Update Steps (ADMM Iterations)

ADMM alternately updates:

$$
\begin{aligned}
\mathbf{x}^{(k+1)} &= \min_{\mathbf{x}} \; f_1(\mathbf{x}) + f_2(\mathbf{x}) \\
\mathbf{z}^{(k+1)} &= \min_{\mathbf{z}} \; g_1(\mathbf{z}) + g_2(\mathbf{z}) \\
\boldsymbol{\gamma}^{(k+1)} &= \boldsymbol{\gamma}^{(k)} + \rho(\mathbf{z}^{(k+1)} + \mathbf{A}\mathbf{x}^{(k+1)} - \mathbf{y})
\end{aligned}
$$

$$\mathbf{x}^{(k+1)} = \min_{\mathbf{x}} \; f_1(\mathbf{x}) + f_2(\mathbf{x}) \tag{6a}$$

$$\mathbf{z}^{(k+1)} = \min_{\mathbf{z}} \; g_1(\mathbf{z}) + g_2(\mathbf{z}) \tag{6b}$$

$$\boldsymbol{\gamma}^{(k+1)} = \boldsymbol{\gamma}^{(k)} + \rho(\mathbf{z}^{(k+1)} + \mathbf{A}\mathbf{x}^{(k+1)} - \mathbf{y}) \tag{6c}$$

with

$$f_1(\mathbf{x}) \triangleq \frac{\rho}{2}||\mathbf{z}^{(k)} + \mathbf{A}\mathbf{x} - \mathbf{y} + \boldsymbol{\gamma}^{(k)}/\rho||_2^2, \quad f_2(\mathbf{x}) \triangleq ||\mathbf{x}||_1,$$

$$g_1(\mathbf{z}) \triangleq \frac{\rho}{2}||\mathbf{z} + \mathbf{A}\mathbf{x}^{(k+1)} - \mathbf{y} + \boldsymbol{\gamma}^{(k)}/\rho||_2^2, \quad g_2(\mathbf{z}) \triangleq \tau||\mathbf{z}||_1. \tag{7}$$

But these equations are not directly solvable. So we use proximal operator for f2(x), g2(z). And linearize f1(x) and g1(z) around x+ and z+

Proximal & Soft-Thresholding Operator

Proximal operator:

$$\mathrm{prox}_{f,t}(x) = \underset{x^+}{\mathrm{argmin}} \ \frac{1}{2t}||x^+ - x||_2^2 + f(x^+),$$

Soft-thresholding (for L1 norm):

$$\mathcal{S}_\alpha(\beta) \triangleq \frac{\max(|\beta| - \alpha, 0)}{\max(|\beta| - \alpha, 0) + \alpha}\beta,$$

Final update equations

$$\mathbf{x}^{(k+1)} = S_{t_x/\rho}\left(\mathbf{x}^{(k)} - t_x/\rho \cdot \nabla f_1(\mathbf{x}^{(k)})\right) \tag{15}$$

$$\mathbf{z}^{(k+1)} = S_{t_z\tau/\rho}\left(\mathbf{z}^{(k)} - t_z/\rho \cdot \nabla g_1(\mathbf{z}^{(k)})\right). \tag{16}$$

Convergence Conditions

Stopping condition based on primal and dual residuals:

$$\mathbf{r}_p^{(k+1)} = \mathbf{A}\mathbf{x}^{(k+1)} + \mathbf{z}^{(k+1)} - \mathbf{y} \tag{17}$$

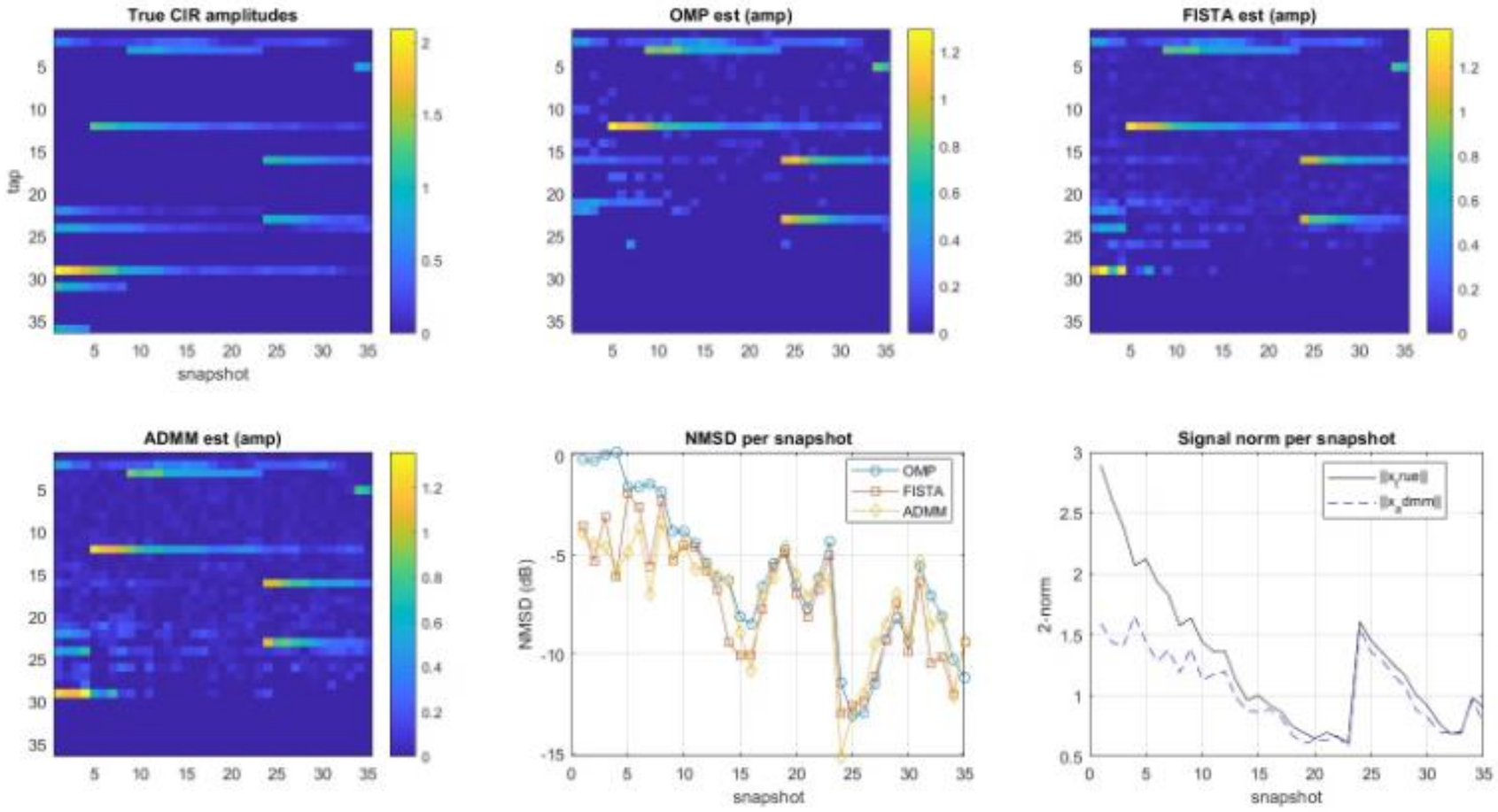
$$\mathbf{r}_d^{(k+1)} = \rho \mathbf{A}^H (\mathbf{r}_p^{(k+1)} - \mathbf{r}_p^{(k)}) - \frac{1}{t_x^{(k+1)}} (\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}). \tag{18}$$

The iteration stops when both the residuals are below tolerance limits

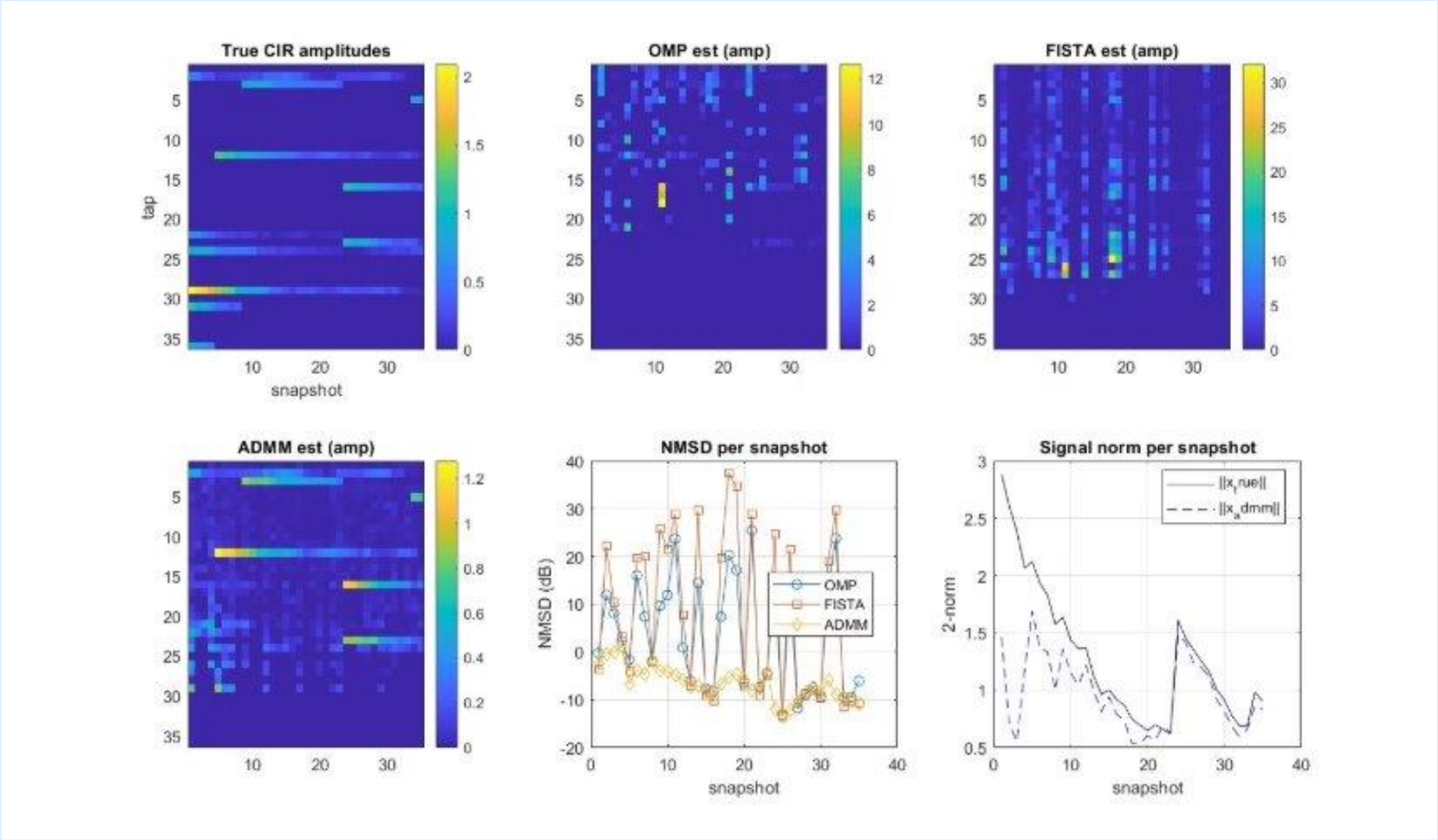
Tolerance is set by us manually

Result

Simulation at SINR=-inf



Simulation at SINR=40



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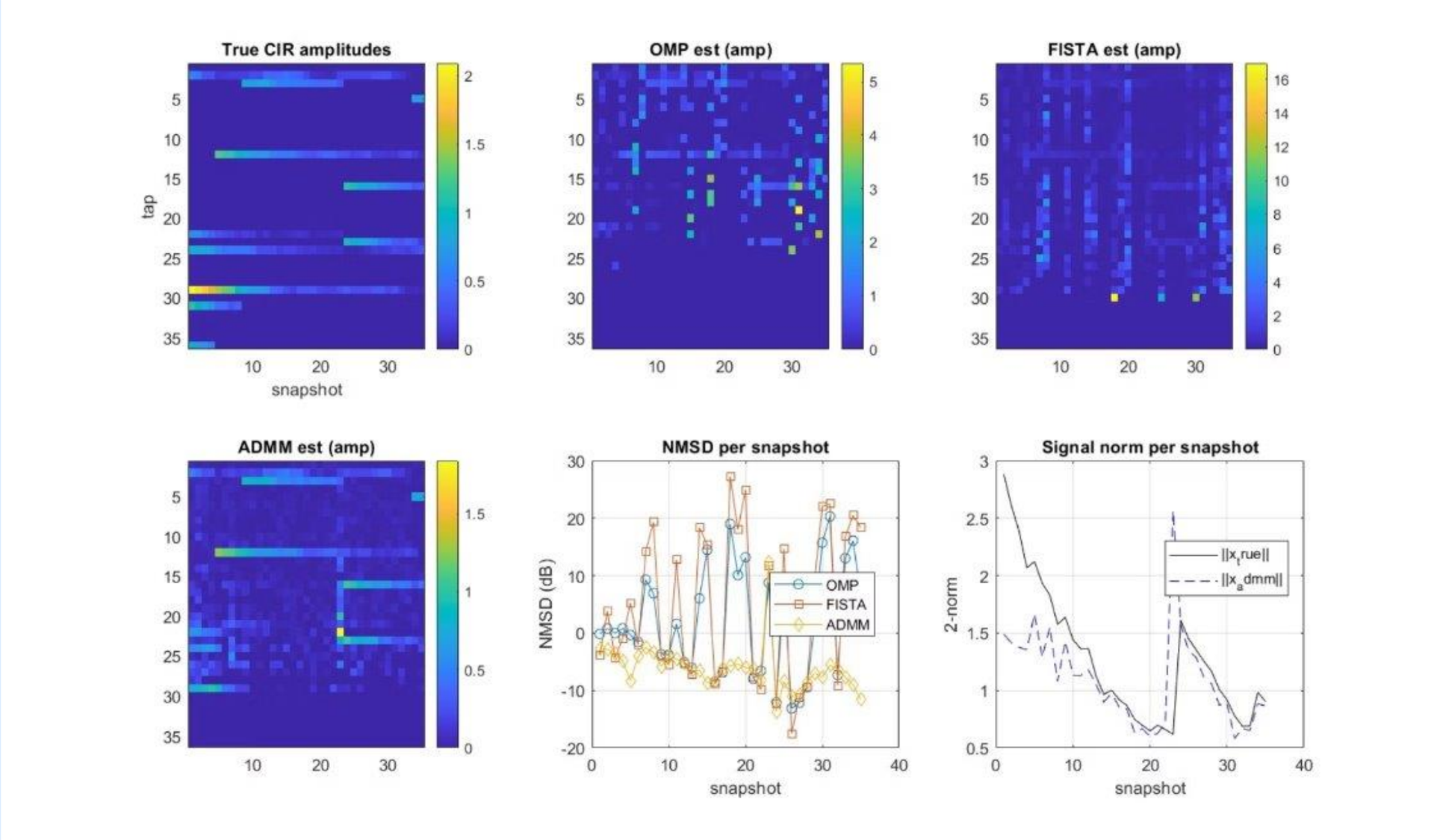
Simulation at SINR=50



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Conclusion

The proposed Robust ADMM algorithm efficiently estimates sparse UWA channels.It maintains accuracy under impulsive noise and converges faster than OMP & FISTA.Simple, robust, and suitable for real-time underwater communication systems.